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Connecting 16- with 4-Membered Rings: Ring-Opening Polymerization of Bio-Based ω-Pentadecalactone with Amino-Alkoxy-Bis(phenolate) Yttrium Initiators and Copolymerization with β-Butyrolactone

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ABSTRACT

Ring-opening polymerization (ROP) of bio-based lactones is an effective way to overcome limited fossil resources. The 16-membered, musk-derived ω-pentadecalactone (PDL) occurs naturally as an ω-hydroxy fatty acid. As PDL is considered a macrolactone, controlled ROP of this *transoid* lactone is mostly obtained *via* enzymatic routes. Herein, amino-alkoxy-bis(phenolate) yttrium amido and alkoxide catalysts are reported for controlled metal-catalyzed ROP of PDL. PDL polymerizations are conducted at elevated temperatures from 60 to 100 °C to produce poly(ω-pentadecalactone) (PPDL) with number-average molecular weights of up to 134.5 kg mol^{-1} and moderate polydispersities (1.5 to 2.1). Utilizing *iso*propanol as a chain transfer agent (CTA) lead to higher initiator efficiencies and low polydispersities while still showing full monomer conversions with molar masses tunable by the amount of CTA. Block copolymerizations of PDL are performed *via* sequential addition. Four-membered, *cisoid*, and rather reluctant, *racemic* β-butyrolactone (BBL), which can be synthesized from carbon dioxide, is chosen as the second block. After determining that PDL has a lower coordination strength to the yttrium center and has to be added as the first monomer, successful connection between PPDL and poly(3-hydroxybutyrate) (PHB) blocks is shown *via* SEC and DOSY NMR spectroscopy. Different ratios of PDL and BBL are chosen to obtain PPDL-*b*-PHB polymers with different block length and molar masses. Powder X-ray diffraction measurements and DSC measurements prove the preservation of crystallinity in the block copolymers.

1. Introduction

The use of petrol-based chemicals is directly coupled to a reduction in fossil resources. Since these resources are limited, the aim of designing bio-derived substitutes from renewable resources is self-evident. One way to produce bio-derived polymers is the ring-opening polymerization (ROP) of naturally occurring lactones. ω-Pentadecalactone (PDL) is a naturally occurring macrocyclic ω-hydroxy fatty acid, present in macrocyclic musk. It is an FDA-approved substance and used for fragrances [1]. It can be extracted from angelica plant root (Angelica archangelica) [2] or musk deer (Moschus) [3]. Polymeric PDL (poly (ω-pentadecalactone) (PPDL)) offers promising characteristics as it is an aliphatic polyester with a chain of 14 methylene groups in its repeating unit, resulting in similar thermal and material properties as fossil-based plastics, e.g. low-density or high-density polyethylene, dependent on its molecular weight [4]. While small lactones are known to polymerize due to a release of ring strain in an enthalpy driven process, macrolactones do not show any internal ring strain [5]. It is assumed that larger rings, such as PDL, polymerize in an entropically driven manner [6]. Kobayashi was the first to publish on enzymatically-catalyzed PDL polymerizations with different lipase enzymes [7]. This was followed by reports from other groups on additional enzymes that successfully catalyze the ROP of PDL [8–10]. When using lipase enzymes in ring-opening polymerization a reverse tendency of ring strains was

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observed as macrolactone polymerization was favored over smaller rings with higher ring strain. Due to the their similarity to fatty acids and hydrophobicity, macrolactones have a stronger recognition of the lipase and are therefore more reactive for enzymatic ROP [1,11–13]. *Transoid* and hydrophobic PDL showed a preferred binding mode, so it can easily be converted to polymer by enzymes [14]. Additionally, the ring strain in smaller lactones is caused by the *cisoid* ester bond. In addition to ineffective, enzymatic ROP of small lactones, another drawback of enzymatic ROP provoked our research on metal-catalyzed ROP of macrolactones: Polymerizations are accompanied by an existence of rapid transesterification reactions which can be intra- and intermolecular. Intramolecular transesterifications form cyclic oligomers, intermolecular ones lead mainly to a broadening in polydispersity both further inhibiting block copolymerizations especially at high monomer conversions [1,15–17]. Therefore, different types of metal-[3,6,18–24] and organo-[6,25–29] catalysts have been reported to produce PPDL in which especially yttrium *iso*propoxide [17,19], yttrium phosphasalen [6], zinc and calcium Schiff base [30] and aluminum-salen [21] catalysts showed promising results in terms of conversions, polydispersities and molecular weights while still requiring high temperatures above 80 °C. For a more comprehensive overview of catalysts for PDL polymerization we refer to Ref. [3]. Feijen *et al.* used yttrium *iso*propoxide as an initiator to produce PPDL with full conversion at 80 °C after 30 min with a molecular weight of 15 kg mol^{-1} ($Đ = 2.1$) [19]. Shuai *et al.* explored these yttrium *iso*propoxide catalysts for industrial-scale production at 100 °C with 97 % yield after 120 min for a monomer to initiator ratio of 800 resulting in molecular weights of up to 50.1 kg mol^{-1} ($Đ = 1.6$) [17]. The group of Duchateau mediated the polymerization by Zn- and Ca-based tridentate Schiff base complexes and benzyl alcohol as initiator at 100 °C. While the zinc Schiff base yielded only 54 % conversion of PDL producing polymers with molecular weights of 73 kg mol^{-1} and an increased polydispersity of 2.3, the calcium Schiff base led to 99 % PDL conversion and polymer molecular weights of 23.8 kg mol^{-1} with the same polydispersity of 2.3 [30]. The same group investigated aluminum-salen complexes. Polymerization at 100 °C in the presence of benzyl alcohol induced full conversion of PDL and PPDL with a molar mass of 41 kg mol^{-1} ($Đ = 2.6$) was obtained [21].

A few examples of random copolymers of PDL mostly with larger lactones can be found in literature [16,28,31–33]. To the best of our knowledge, except from a few block copolymers synthesized *via* sequential addition of PDL with ε-caprolactone (CL) [15,22,34], δ-valerolactone (VL) [34] or lactide (LA) [15,35], PDL block copolymers with smaller lactones have not successfully been synthesized so far. Commonly used ring sizes for metal-catalyzed ROP have been shown to range from four-membered ring β-butyrolactone (BBL) [36–39] to bigger lactones such as lactide [40–42] and ε-caprolactone [20,43–45].

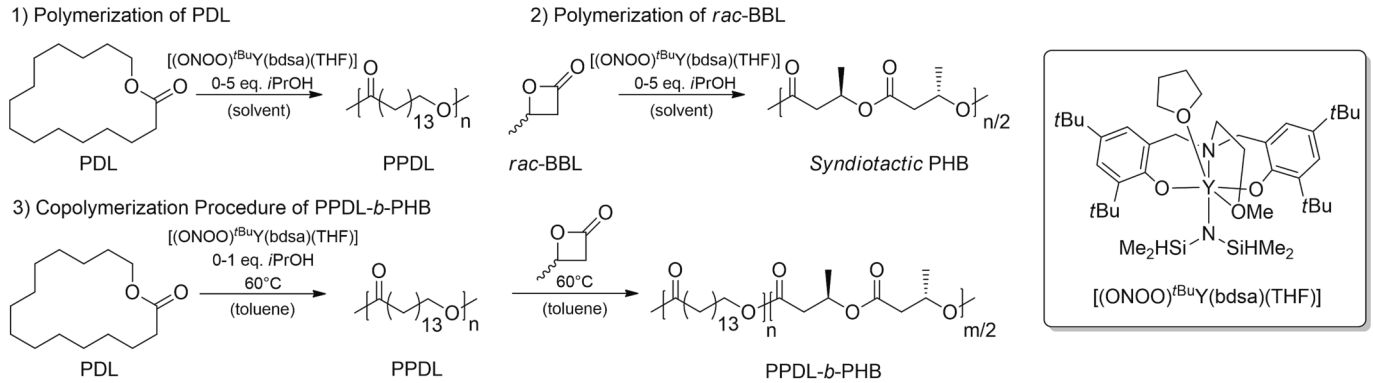
Four-membered BBL can be considered as a sustainable monomer, as it can be produced from carbon dioxide [46–52]. The driving force of BBL polymerizations is the relief of ring strain by opening the *cisoid* lactone ester bond, however, only a small number of metal-catalysts are able to polymerize this rather reluctant monomer to obtain poly(3-hydroxybutyrate) (PHB) [13,53,54].

Coates *et al.* reported on zinc alkoxide initiators for the ROP of *racemic* BBL *via* coordination-insertion mechanism for low polydispersities resulting in mainly atactic PHB, which does not have sufficient properties for any application [55]. The group of Carpentier focused on the *syndiospecific*, lanthanide-mediated ROP of *racemic* BBL using bis(phenolate) catalysts, namely an amino-alkoxy-bis(phenolate) yttrium amido complex [(ONOO)^{tBu}Y(bdsa)(THF)] (Scheme 1), with ONOO^{tBu} = 6,6′-(((2-methoxyethyl)azane-diyl)-bis(methylene))-bis(2,4-di-*tert*-butylphenolate), bdsa = bis(dimethylsilyl)amide and THF = tetrahydrofuran [37,38]. Compared to other catalyst systems, in particular these lanthanide complexes supported by single-site dianionic aminoalkoxy-bis(phenolate) ligands provide the opportunity of a rapid and controlled ROP, resulting in high molar mass PHB under mild conditions outperforming most of the other commonly used, active, and stereoselective catalysts for BBL polymerization [56,57]. For example, *in situ* generated "Y(O*i*Pr)$_3$" polymerized atactic PHB ($P_r = 0.51$) thus confirming the necessity of bis(phenolate) ligands for the control of tacticity during BBL polymerization [38].

Rieger *et al.* used similar catalysts with heteroaromatic initiators for copolymerization of BBL in order to further tune material properties of the semi-crystalline, *syndiotactic* PHB, due to its attractive properties as a replacing material for poly(propylene) and other fossil-based thermoplastics [36,58].

Herein, the [(ONOO)^{tBu}Y(bdsa)(THF)] catalyst is shown to be active in PDL polymerization *via* ROP-mechanism in a controlled manner (Scheme 1.1). PDL polymerization was further optimized by using *iso*propanol (*i*PrOH) as a chain transfer agent (CTA) leading to an *in situ* generated bis(phenolate) yttrium alkoxide initiator for immortal ROP (iROP). Addition of CTA offered tunability of the molecular weights while simultaneously lowering the polydispersity. Moreover, the use of an alkoxide initiator enhanced the initiator efficiencies and led to higher conversions of PDL.

The possibility of combining semi-crystalline PHB and PPDL in AB block copolymers (PPDL-*b*-PHB) was demonstrated *via* sequential addition of *transoid* PDL and *cisoid* BBL catalyzed by amino-alkoxy-bis(phenolate) yttrium amido and alkoxide complexes (Scheme 1.3). Block copolymer formation was proven *via* NMR spectroscopy and size-exclusion chromatography (SEC). Differential scanning calorimetry (DSC), thermogravimetric analysis (TGA) and powder X-ray diffraction (PXRD) were used to study the thermal properties of the polymers and to prove the preservation of crystallinity of both polymeric blocks.



Scheme 1. 1) Synthesis of PPDL *via* ring-opening polymerization of PDL. 2) Polymerization of *rac*-BBL to obtain *syndiotactic* PHB. 3) Copolymerization procedure *via* sequential addition of PDL and *rac*-BBL to synthesize PPDL-*b*-PHB. All reactions were performed with a [(ONOO)^{tBu}Y(THF)] amido catalyst with or without *iso*propanol for *in situ* preparation of the [(ONOO)^{tBu}Y(THF)] alkoxide catalyst.

2. Materials and Methods

2.1. Materials

All chemicals were purchased from *TCI chemicals* or *Sigma-Aldrich* and were used as received, unless otherwise stated. Diethyl ether, dichloromethane, pentane, thf and toluene were dried with a solvent purification system (*MBraun*) and stored inside an argon-filled glovebox (*MBraun EcoLab*) over molecular sieves. All reactions were performed under argon atmosphere using Schlenk technique or in a glovebox. Prior to use, ω-pentadecalactone (PDL) was dried under vacuum using a Schlenk line setup. *Racemic* β-butyrolactone (BBL) was stirred over barium oxide (BaO). After separating the BBL from BaO *via* centrifugation, BBL was dried over calcium hydride and distilled prior to use.

The synthesis of the amino-alkoxy-bis(phenolate) yttrium amido catalyst [(ONOO)$^{t\mathrm{Bu}}$Y(bdsa)(THF)], was performed according to literature [59]. It involved the synthesis of an yttrium-bdsa-precursor [60] (Y(bdsa)$_3$(THF)$_2$, bdsa = bis(dimethylsilyl)amide), and the 2-methoxyethylamino-bis(phenolate) ligand [61].

2.2. Methods

Homopolymerizationprocedure of ω-pentadecalactone and *racemic* β-butyrolactone with [(ONOO) $^{t\mathrm{Bu}}$Y(bdsa)(THF)] catalyst.

The respective amount of ω-pentadecalactone was dissolved in 1.5 ml toluene, stirred and pre-heated to the respective temperature. 19.94 mg of [(ONOO)$^{t\mathrm{Bu}}$Y(bdsa)(THF)] catalyst (24.9 μmol, 1.0 eq.) was dissolved in 0.5 ml toluene and stirred at the respective temperature. After five minutes, the catalyst solution was added to the stirred monomer solution in one portion. The reaction mixture was stirred for the respective time and at the given temperature until being quenched by the addition of 1.5 ml of wet deuterated chloroform. A 0.5 ml aliquot was taken from the solution for the determination of the conversion *via* ^{1}H NMR spectroscopy. The residual reaction solution was precipitated from 100 ml methanol. The polymer was air-dried after filtration. Relative molar masses and polydispersities were determined *via* size exclusion chromatography (SEC) in chloroform using polystyrene standards.

BBL homopolymers were produced according to literature protocols [36].

Kinetic measurements of the homopolymerization reactions with [(ONOO)$^{t\mathrm{Bu}}$Y(bdsa)(THF)] catalyst.

An aliquot method was used for kinetic measurements. ω-Pentadecalactone (2.49 mmol, 50 eq., 598.55 mg) was dissolved in 3 ml of toluene and stirred in a heating block at the respective temperature. [(ONOO)$^{t\mathrm{Bu}}$ Y(bdsa)(THF)] catalyst (49.8 μmol, 1.0 eq., 39.78 mg) was dissolved in 1 ml toluene, the desired amount of *iso*propanol was added, the mixture stirred for five minutes at the respective temperature and added to the stirred monomer solution in the heating block in one portion.

For BBL polymerization, [(ONOO) $^{t\mathrm{Bu}}$Y(bdsa)(THF)] catalyst (49.8 μmol, 1.0 eq., 39.78 mg) was dissolved in 4 ml toluene, stirred and pre-heated in the heating block and the BBL monomer (4.98 mmol, 100 eq., 428.73 mg) was added in one-shot.

Aliquots of 100 μL were taken at regular time intervals and quenched by addition of wet deuterated chloroform. The conversion of every aliquot was determined *via* ^{1}H NMR spectroscopy, the relative molar masses and polydispersities of each sample were measured relative to polystyrene standards using SEC with chloroform as eluent.

Copolymerization procedure with [(ONOO)$^{t\mathrm{Bu}}$Y(bdsa)(THF)] catalyst.

Copolymers were produced *via* sequential addition of monomers. The respective amount of ω-pentadecalactone, as the first block, was dissolved in 1.5 ml of toluene and stirred in a heating block at the respective temperature. The catalyst solution (24.9 μmol catalyst, 1.0 eq. in 0.5 ml toluene) was added in one portion to the monomer solution. The reaction mixture was stirred for the given time interval at the chosen temperature. A 100 μL aliquot was then taken and quenched by addition of 400 μL wet deuterated chloroform. The respective amount BBL was added and the reaction mixture was further stirred. A 100 μL aliquot of the second block was taken and quenched by wet $CDCl_3$. Conversions, relative molar mass and polydispersities of the aliquots were determined by ^{1}H NMR spectroscopy and SEC in chloroform relative to polystyrene standards. The reaction mixture was dissolved in chloroform, precipitated from methanol and air-dried after filtration. The dried block copolymer was used for further analysis.

Polymerizations with a chain transfer agent for *in situ* preparation of [(ONOO)$^{t\mathrm{Bu}}$Y(*i*PrO)(THF)] catalyst.

Polymerizations with a chain transfer agent were performed as the homopolymerizations described above but included an extra step. The monomer was dissolved in 1.5 ml solvent, stirred and pre-heated. In another vial, 0.5 ml solvent were added to respective amount of catalyst. The respective equivalents of anhydrous *iso*propanol were added to the [(ONOO)$^{t\mathrm{Bu}}$Y(bdsa)(THF)] complex solution which was stirred for further 5 min. The catalyst solution was added to the monomer solution to start the polymerization. Addition of 1.5 ml wet deuterated chloroform quenched the polymerization at the respective time point. An aliquot was taken for ^{1}H NMR measurements to determine conversion and the reaction mixture was precipitated from 100 ml methanol, followed by air drying after filtration.

2.3. Analysis

Nuclear magnetic resonance (NMR) spectra were recorded using a *Bruker* Avance III 400, *Bruker* Avance 500, or *Bruker* Avance III HD 700 NMR spectrometer with a 400, 500, or 700 MHz magnet. Chemical shifts, unless otherwise stated in ppm, were recorded in relation to the residual proton signal of the deuterated solvent in which the substance was dissolved.

The size exclusion chromatography (SEC) was performed with chloroform on a SEC setup containing a *PSS* SDV 5 μm 8*50 mm guard column with three *PSS* SDV 100,000 Å 5 μm 8*50 mm columns heated at 40 °C. The columns were connected with the refractive index (RI) detector (*Agilent* 1200 Series G1362A). Samples had concentrations between 2 and 4 mg/ml. Relative molar masses were determined using polystyrene standards.

Differential scanning calorimetry (DSC) was measured on a *PerkinElmer* DSC 4000 with samples of 5–10 mg. It was measured in an endo up mode and a heating/cooling rate of 10 K/min in three cycles. The second cycle was used for evaluation.

For thermogravimetric analysis (TGA) measurements, samples of 15–20 mg were heated to 900 °C with 10 °C/min using a Thermogravimetric Analyzer TGA 4000 from *PerkinElmer*.

Gas chromatography–mass spectrometry (GC–MS) was performed with a 5977D GC system by *Agilent*. The sample was dissolved with a concentration of 2 mg/ml in acetone.

Electrospray ionization coupled with mass spectrometry (ESI-MS) measurements were carried out in acetonitrile with a *Varian* 500-MS spectrometer or a micrOTOF-Q *Bruker* Daltonics in positive ionization mode.

Positive ion matrix-assisted laser desorption ionization time-of-flight (MALDI-TOF) measurements were performed with a *Bruker* Autoflex III with smart beam in reflector mode. Samples were prepared from a tetrahydrofuran solution. The matrix (2,5-dihydrobenzoeic acid) (5 mg/mL in thf) was mixed with the sample (5 mg/mL in thf) and sodium trifluoromethanesulfonate (0.1 M in 90 % acetone, 10 % water) in a ratio of 2:1:2.

Powder X-ray diffraction (PXRD) was measured with a *Stoe* IPDS diffractometer in transmission mode with Cu-K$_\alpha$ radiation with a wavelength of 154.05 pm coupled to a HyPix3000 detector. The baseline correction was done manually. All block copolymer data were normalized to the highest intensity reflex of the measurement of pure poly

(ω-pentadecalactone. The measurement of poly(3-hydroxybutyrate) was used for comparison reasons and was normalized to the highest PHB reflex of the block copolymer with the highest PHB content (Table 2, Entry 5).

3. Results and Discussion

3.1. Homopolymerization of ω-Pentadecalactone

Initially, pentadecalactone oligomers were prepared with [(ONOO)^{tBu}Y(bdsa)(THF)] to perform end-group analysis *via* ESI-MS and MALDI-TOF measurements. For ESI-MS analysis, signals for $m/z = (n \times M_{PDL} + Na)^+$ and $m/z = (n \times M_{PDL} + K)^+$ were observed for repeating units $n = 2, 3, 4$ with Na^+ and K^+ operating as charge carrier (Figure S24). The absence of bisdimethylsilyamide end-groups can be explained by fragmentation due to the ESI-MS measurement as already obtained elsewhere [36,39]. Time-resolved ^{1}H NMR spectra in previous mechanistic investigations showed a decrease of the coordinated bdsa signal which was related to bdsa acting as chain initiator [39]. ESI-MS measurements were also performed with PDL reacted with [(ONOO)^{t}-^{Bu}Y(bdsa)(THF)] catalyst and *i*PrOH (Figure S25). *Iso*propanol was introduced for *in situ* preparation of an [(ONOO)^{tBu}Y(alkoxide)(THF)] catalyst. Signals for $m/z = (n \times M_{PDL} + H)^+$, $m/z = (n \times M_{PDL} + Na)^+$, and as well for $m/z = (n \times M_{PDL} + M_{iPrOH} + H)^+$, and $m/z = (n \times M_{PDL} + M_{iPrOH} + Na)^+$ could be detected for $n = 1, 2, 3$ in accordance with initiation by an *iso*propoxide initiator. Additional MALDI-TOF measurements support this outcome and increase the detectable repeating units. When using [(ONOO)^{tBu}Y(bdsa)(THF)] oligomers with $n = 3$–8 and $m/z = (n \times M_{PDL} + Na)^+$ (Figure S26) were observed. When using [(ONOO)^{tBu}Y(^{i}PrO)(THF)], oligo(PDL) with $n = 3$–6 according to ($m/z = (n \times M_{PDL} + M_{iPrOH} + Na)^+$) were visible (Figure S27). These findings suggest that *iso*propoxide can serve as an initiator and cyclic species are not only formed.

After oligomerization of PDL, PDL was polymerized with a catalyst concentration of 24.9 μmol in 2 ml toluene over night to evaluate general activities and initiator efficiencies at different temperatures and catalyst loadings (Table 1). Reactions at room temperature showed to be barely sufficient (Table 1, Entry 1) as low PDL conversions (<38 %) and PPDL molecular weights (15.1 kg mol^{-1}) were observed accompanied by high polydispersities ($Đ = 3.1$). High polydispersities can be caused by side reactions, such as transesterifications or by a slow and non-simultaneous initiation caused by an inefficient initiation process at room temperature.

Reactions at 60 °C showed successful polymerization of PDL (Table 1, Entries 2 and 5–7). Polymerizing PDL with a target DP of 25 (DP = degree of polymerization) produced the lowest polydispersity of 1.5 (Table 1, Entry 2). By increasing the equivalents of [Cat]/[PDL] to 50 and 100 PDL conversions were still high (>94 %), but polydispersities of PPDL were increased to 1.9 and 2.0 with molecular weights of 42.3 and 98.0 kg mol^{-1}, respectively (Table 1, Entries 5 and 6). High molecular weights of up to 134.5 kg mol^{-1} could be reached by using 200 equivalents of PDL, maintaining a moderate polydispersity of 2.1 with a PDL conversion of 85 % (Table 1, Entry 7). Neat reactions were performed at 60 °C to increase initiator efficiencies (Table 1, Entry 3). Polymerization results showed full PDL conversion and polymers with only half the molecular weight of a comparable reaction in toluene were obtained indicating an increased initiator efficiency. However, polydispersities were high ($Đ = 4.2$) and the polymer was not processible for further analysis indicating an uncontrolled reaction under bulk conditions. ROP of PDL with DP 50 in tetrahydrofuran (THF) showed 95 % PDL conversion and polymers with a molecular weight of 49.4 kg mol^{-1} ($Đ = 1.8$) were produced indicating a slightly lower initiator efficiency than in toluene (0.23 vs. 0.27) (Table 1, Entry 4 and 5).

To examine the influence of temperature on polymerization activity and polymer characteristics, polymerizations with a DP of 50 were also conducted at 80 and 100 °C (Table 1, Entries 8 and 9). Even with similar PDL conversions compared to the 60 °C polymerization (94 % for 80 °C, 95 % for 100 °C), molecular weights were slightly increased to 47.7 kg mol^{-1} at 80 °C and 57.3 kg mol^{-1} at 100 °C, showing that the catalyst has an insignificantly lower initiator efficiency (60 °C: 0.27, 80 °C: 0.24, 100 °C: 0.20) at elevated temperatures probably due to slight degradation of the catalyst. We do not assume a higher proportion of side reactions since the polydispersities are even lower than for the 60 °C reaction indicating a controlled polymerization.

When *i*PrOH was utilized to *in situ* convert the bdsa complex into the correspondent [(ONOO)^{tBu}Y(*i*PrO)(THF)] catalyst an improvement PDL polymerization was observed (Table 1, Entry 10–14). Weak initiation groups such as bdsa can react with impurities, e.g. water, to form new initiating groups. Consequently, the initiation rate is slower than the propagation rate leading to broadening of the molecular weight distribution. *Iso*propoxide groups are most likely to be directly inserted into the acyl-oxygen bond because they mimic the propagating groups of the presumed active species facilitating narrower molecular weight distributions [40,62]. Almost full PDL conversion was observed when using the *in situ* generated [(ONOO)^{tBu}Y(*i*PrO)(THF)] and polymers with lower polydispersities were isolated. Initiator efficiencies could be raised up to 0.7 (1.0 eq. *i*PrOH; Table 1, Entry 10).

As stated in literature, PPDL is semi-crystalline with a glass transition temperature of −27 °C and a melting temperature of around 100 °C. Research has shown that these temperatures correlate to the molecular weight [4]. Values for glass transition temperatures could not be detected with the DSC measurements due to the high crystallinity of PPDL. The values for melting temperatures were similar for all 60 °C reactions with a value of 93 to 97 °C, increasing slightly for higher target DPs of 100 and 200, because of higher molecular weight PPDL. PPDL produced at 80 and 100 °C with a DP of 50 showed the same T_m of 94 °C as for the 60 °C reaction, indicating that the polymerization temperature does not influence thermal properties and crystallinity of the polymers. In similarity, the use of different initiators did not affect the melting temperatures of the obtained polymers.

Kinetic measurements were used to study the influence of temperature on the activity of the two employed catalysts in PDL polymerization at 60 °C, 80 °C and 100 °C (Fig. 1). SEC traces from these aliquots picture

Table 1
Selected data for PDL polymerizations.[a]

Entry	[Y]: [PDL]: [*i*PrOH]	*T* [°C]	Conv. [%][b]	$M_{n,calc}$ [kg mol^{-1}][c]	$M_{n,SEC}$ [kg mol^{-1}][d]	$Đ$[d]	T_m [°C][e]
1	1:50:-	RT	38	4.6	15.1	3.1	94
2	1:25:-	60	>99	6.0	22.0	1.5	93
3[f]	1:50:-	60	>99	11.9	22.4	4.2	n.d.
4[g]	1:50:-	60	95	11.5	49.4	1.8	94
5	1:50:-	60	94	11.3	42.3	1.9	94
6	1:100:-	60	96	23.1	98.0	2.0	97
7	1:200:-	60	85	39.9	134.5	2.1	96
8	1:50:-	80	94	11.3	47.7	1.7	94
9	1:50:-	100	95	11.5	57.3	1.7	94
10	1:50:1	60	>99	12.0	17.3	1.6	94
11	1:50:3	60	>99	4.0	12.6	1.6	95
12	1:50:5	60	>99	2.4	8.2	1.6	93
13	1:100:1	60	>99	24.0	24.8	1.7	96
14	1:200:1	60	>99	48.0	48.8	1.8	97

[a] Reactions performed for 18 h with 24.9 μmol [(ONOO)^{tBu}Y(bdsa)(THF)] in 2 ml toluene except from the Entry 3 and 4; [b]Conversion determined *via* ^{1}H NMR spectroscopy by integration of the -O-CH_2- groups of PDL and PPDL; [c]$M_{n,calc}$ calculated by $M_{n,calc}$=$M_{monomer}$*([PDL]/[Y])*conversion, when *i*PrOH is used then the caluclation is as follows: $M_{n,calc}$=($M_{monomer}$*([PDL]/[Y])*conversion)* 1/[*i*PrOH]; [d]Number-average molecular weight as measured *via* SEC in chloroform at 40 °C relative to polystyrene standards. Initiator efficiency calculated via $M_{n,calc}/M_{n,SEC}$ by using relative obtained molar masses. Therefore, efficiencies are only calculated for comparison reasons and are not absolute values; [e]Measured via DSC with a heating rate of 10 K min^{-1} (second heat cycle); [f]Neat reaction without solvent; [g]Polymerization in 2 ml THF.

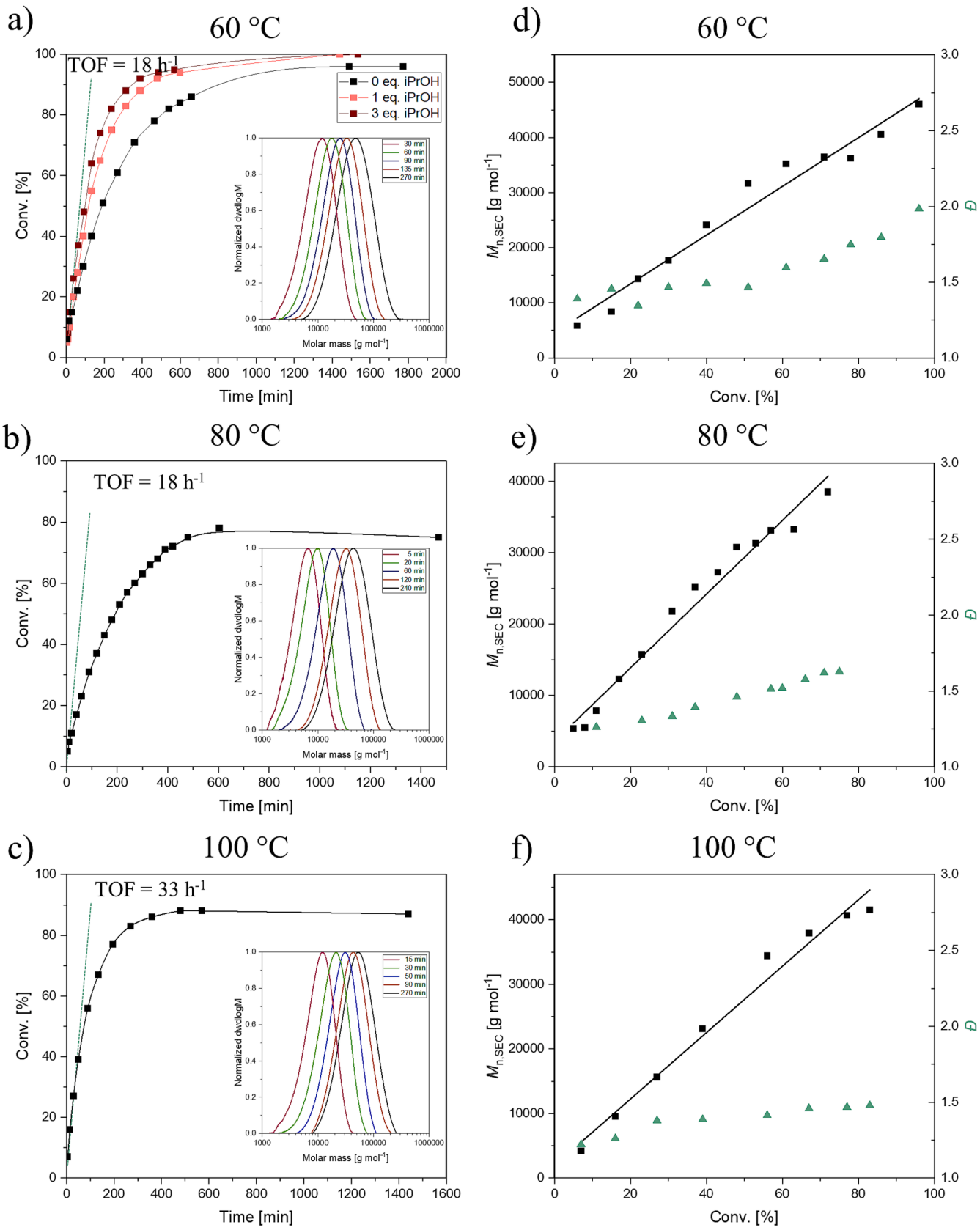


Fig. 1. Kinetic measurements of PDL polymerization at different temperatures. Left: Conversion, determined by ^{1}H NMR measurements, over reaction time and growth of number-average molar mass with time at 60 °C (a), 80 °C (b) and 100 °C (c). Right: Dependency of number-average molar mass (M_n) and polydispersity (*Đ*) on the conversion for 60 °C (d), 80 °C (e) and 100 °C (f).

the growth of the number-average molar masses with ongoing reaction time for these temperatures. The TOF using [(ONOO)^{tBu}Y(bdsa)(THF)] was determined to be 18 h^{-1} for 60 °C and 80 °C. Reaction at 100 °C showed a TOF of 33 h^{-1}. In comparison to literature, an improvement in activity can be obtained from these investigated systems. An yttrium phosphasalen catalyst at room temperature was active with a TOF of 25 h^{-1}. This catalyst reached a TOF of 126,000 h^{-1} in a bulk polymerization at 100 °C, but with a high polydispersity ($Đ = 2.8$) [6]. Using 1,5,7-triazabicyclo[4.4.0]dec-5-ene (TBD) in combination with benzyl alcohol (BnOH) as initiator in a PDL bulk polymerization at 100 °C was reported to give TOFs of up to 17 h^{-1} [28], while a $Zn(C_6F_5)_2$/1,8-diazabicyclo[5.4.0]undec-7-ene (DBU) Lewis pair at 110 °C exhibited a TOF of 15 h^{-1} [15]. One phosphazene superbase in combination with 3-phenyl-1-propanol as initiator reached the TOF 124 h^{-1} ($Đ = 2.7$) at 80 °C. Decreasing the concentration in solution for the ROP led to a TOF of 92 h^{-1} while also lowering the polydispersity ($Đ = 1.8$) [27].

Studies were also conducted at 60 °C with 1 and 3 equivalents of *iso*propanol (Fig. 1a). Both reactions reached almost full conversion after 10 h. Whereas the respective amido catalyst only facilitated a PDL conversion of 96 % at the same temperature, alkoxide initiation led to full conversion after 18 h. Polymerizations with the [(ONOO)^{tBu}Y(bdsa)(THF)] catalyst at 80 and 100 °C enabled a maximum conversion of PDL of 78 and 88 %. During polymerization without taking aliquots monomer conversions of 94 and 95 % for 80 and 100 °C were reached. The reactions at elevated temperatures, above 60 °C, had to be performed outside of the glovebox in screw cap vials with a polypropylene septum. Especially during kinetic measurements, contaminations by water and air while taking aliquots could impact the polymerization results which is reflected by lower PDL conversions and PPDL molar masses (80 °C: 38.5 kg mol^{-1} and 100 °C: 41.6 kg mol^{-1}) in comparison to polymerizations in a sealed vial (Table 1, Entry 8 and 9). For reactions that were performed in the glovebox at 60 °C in a sealed vial, similar results for standard polymerizations or kinetic measurements were obtained (46 kg mol^{-1} vs. 42.3 kg mol^{-1} (Table 1, Entry 5)). When plotting the molar mass against the conversion, a linear growth was observed for all temperatures indicating a living-type polymerization setup. However, an increase in polydispersity and a slight discrepancy in molar masses was observed during the 60 °C reactions especially at high PDL conversions, probably due to a longer reaction time, facilitating more side, transesterification and termination reactions in comparison to reactions at 80 and 100 °C. For these temperatures, molar mass distributions stayed moderate throughout the whole polymerization timespan. Therefore, PDL polymerizations were stopped at a point with a high number-average weight and the lowest possible $Đ$ when performing chain extension studies with a second monomer. Polymerizations of PDL at 60 °C were discontinued after reaching the plateau region after 18 h or even earlier.

3.2. Block Copolymers of PPDL with PHB

After successful homopolymerization of PDL, the same bis(phenolate) amido and alkoxide catalysts were used for BBL homopolymerization to screen possible reaction conditions for block copolymerization of BBL with PDL (Table S1). BBL was polymerized at room temperature (Table S1, Entries 1–3) and 60 °C (Table S1, Entries 4 and 5). Polymerizations at room temperature with [(ONOO)^{tBu}Y(bdsa)(THF)] led to higher molecular weights (23 kg mol^{-1}) and lower polydispersities ($Đ = 1.7$) compared to reactions performed at 60 °C ($M_n = 7$ kg mol^{-1}, $Đ = 2.1$) showing that an increased temperature leads to both, a higher initiator efficiency and more side/transesterification/termination reactions. The use of *iso*propanol for *in situ* preparation of the respective alkoxide catalyst further increased the control over the reaction, even at an elevated temperature of 60 °C, and PHB with very low polydispersity of 1.1–1.2 was obtained (Table S1, Entry 6–8). Addition of more than 5 equivalents of alcohol were reported to interrupt catalytic activity of *rac*-BBL [63]. In this research, 1–5 equivalents of *i*PrOH were successfully introduced as chain transfer agent to obtain a controlled, immortal polymerization accompanied by lower molar masses controlled by the amount of chain transfer agent and due to a better initiator efficiency. Melting temperatures, determined by DSC, of PHB from polymerization in toluene at room temperature were higher than from polymers produced at higher temperatures which is induced by a higher *syndiotacticity* leading also to a higher crystallinity. PHB produced at room temperature showed a T_m of 122 °C (Table S1, Entry 1) while a 60 °C polymerization temperature led to polymers with melting temperatures around 109 °C (Table S1, Entry 4). These findings are explained by the chain-end control mechanism during polymerization in which stereoregulating ability and thus crystallinity drops with increasing temperatures [38,62,64]. An increase in tacticity of PHB was also observed when changing the solvent to dichloromethane (DCM) (Table S1, Entry 2), due to the superior solvation characteristics of DCM [39]. A P_r value of 0.83 caused a high melting temperature of 140 °C. Aliquots of BBL polymerization at 60 °C in toluene were taken after full conversion was reached to prove the absence of transesterification reactions shown by a constant molar mass and polydispersity even after three hours of full conversion of BBL (Figure S28-S29). Previous research on BBL polymerization at an elevated temperature of 100 °C had shown that the [(ONOO)^{tBu}Y(bdsa)(THF)] was still catalytically active, but polymers had higher polydispersities attributed to the higher chance of side reactions at elevated temperatures. Moreover, these PHB samples were less *syndiotactic* [36]. PHB obtained from immortal polymerization showed higher melting temperatures ($T_m = 131–133$ °C) and *syndiotactic* fractions in comparison to polymerization with the amide initiator when similar reaction conditions were used (room temperature, toluene as solvent, [Y]:[M] = 1:100) indicating a higher crystallinity of the polymers. These results for BBL polymerization are consistent with the literature for BBL polymerization with the [(ONOO)^{tBu}Y(bdsa)(THF)] catalyst and further complement those [36–39].

Block copolymers were synthesized by sequential addition of monomers PDL and BBL at an optimal temperature of 60 °C in which both monomers could successfully be polymerized with satisfactory results. First copolymerization experiments in which a mixture of both monomers (10 equivalents each) was added to the catalyst, revealed that BBL had a higher coordination strength to the catalyst and polymerized first, whereas PDL showed no conversion (Figure S16). On basis of these findings, PDL had to be polymerized as the first block. After full conversion and before an increase in polydispersity (see part 3.1), BBL was added for chain extension to produce PPDL-*b*-PHB block copolymers.

AB block copolymers with different compositions and molar masses were synthesized (Table 2) with either[(ONOO)^{tBu}Y(bdsa)(THF)] or *in situ* prepared [(ONOO)^{tBu}Y(*i*PrO)(THF)] in toluene at 60 °C. PDL blocks showed similar results as PDL homopolymerizations concerning conversions, molecular weights and molar mass distributions. BBL conversions reached full conversion when using the alkoxide initiator and nearly full conversion when up to 86 equivalents of monomer and amide initiators were used (Table 2, Entry 1–3 & 6–8). For amide complexes, BBL conversions reached 84 % when more than 100 eq. of BBL were used in combination with 50 eq. of PDL (Table 2, Entries 4–5).

^{1}H NMR spectra were used to calculate the ratio of PPDL/PHB (Fig. 2, bottom), based on spectra of the two homopolymers (Fig. 2 top and middle). The PPDL-*b*-PHB NMR spectrum contained all signals of the two single homopolymers giving first evidence of successful block copolymerization. DOSY NMR studies were performed to prove the linkage of PPDL and PHB blocks. DOSY NMR spectra of PPDL-*b*-PHB with different compositions (PPDL/PHB = 49/51 (Figure S19), PPDL/PHB = 13/87 (Figure S20)) showed only one set of signals and thus only one diffusion coefficient attributed to PPDL-*b*-PHB block copolymers. In contrast, the DOSY NMR of a blend of PPDL and PHB (Figure S23), contained two sets of signals, one for each homopolymer species. The polymer with PPDL/PHB = 35/65 had an additional species of PHB homopolymer besides the PPDL-*b*-PHB in its DOSY NMR spectrum (Figure S21), even if it was not the polymer with the highest PHB

Table 2
Selected data for PPDL-*b*-PHB block copolymerizations.[a]

Entry	[Y]:[PDL]: [BBL]: [*i*PrOH]	$Time_{PDL}$ [h]	$Conv._{PDL}$ [%][b]	$M_{n,PPDL,SEC}$ [kg mol^{-1}][c]	$Đ_{PPDL}$[c]	$Time_{BBL}$ [h]	$Conv._{BBL}$ [%][b]	P_r [%][d]	$M_{n,AB,SEC}$ [kg mol^{-1}][c]	$Đ_{AB}$[c]	M_{PPDL}/ M_{PHB}[e]
1	1:25: 86	15.03	82	25.3	1.4	6.20	92	74	29.9	1.4	20/ 80
2	1:50: 28	5.72	66	47.5	1.7	18.07	>99	77	60.5	1.7	61/ 39
3	1:50: 52	19.82	94	37.5	1.6	7.17	98	78	39.0	1.7	49/ 51
4	1:50: 106	15.05	91	41.2	1.7	7.12	84	74	50.7	1.7	35/ 65
5	1:51: 181	6.78	70	31.0	1.6	17.22	84	75	55.7	1.4	13/ 87
6	1:25: 114:1	8.68	78	11.6	1.5	14.87	>99	76	18.6	1.3	15/ 85
7	1:50: 25:1	8.78	90	15.7	1.6	14.82	>99	n.d.	17.0	1.6	71/ 29
8	1:50: 100:1	8.72	89	19.3	1.6	14.73	>99	76	22.0	1.5	32/ 68

[a] All polymerizations were conducted with 24.9 μmol [(ONOO)^{tBu}Y(bdsa)(THF)] in 2 ml of toluene at 60 °C; [b]Determined *via* ^{1}H NMR spectroscopy by integration of the -O-CH_2- signal of PDL and PPDL, or integration of the -O-C*H*- signal of BBL and PHB; [c]Measured *via* SEC in chloroform at 40 °C relative to polystyrene standards; [d]Probability of *racemic* linkages between single monomer units, determined *via* the carbonyl region of ^{13}C NMR analysis; [e]Calculated *via* ^{1}H NMR spectroscopy as depicted in Figure S17.

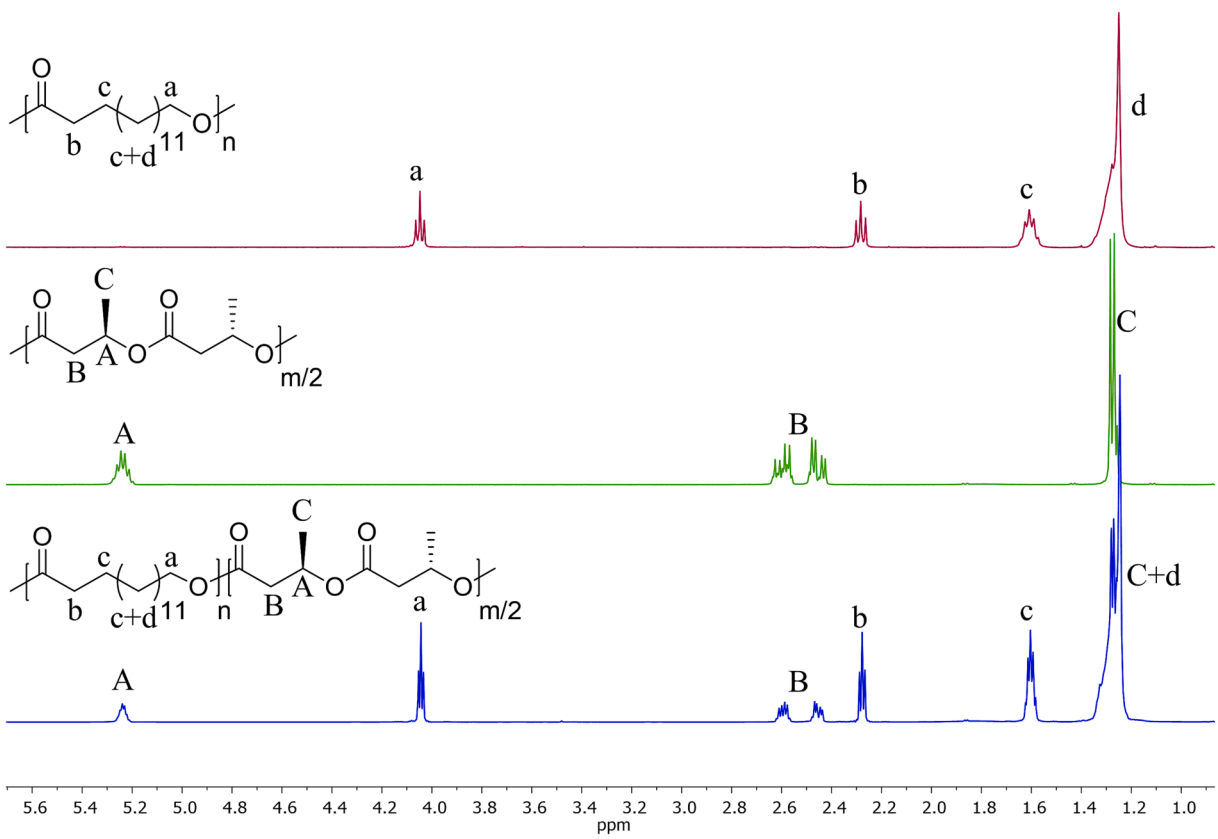


Fig. 2. ^{1}H NMR measurements of PPDL (top, pink) (Table 1, Entry 5, Figure S12, 500 MHz, $CDCl_3$); PHB (middle, green) (Table S1, Entry 2, Figure S6, 500 MHz, $CDCl_3$); PPDL-*b*-PHB (bottom, blue) (Table 2, Entry 3, PPDL/PHB = 49/51, Figure S17, 700 MHz, $CDCl_3$).

content. This was also observed for the block copolymers with similar composition synthesized with *i*PrOH as chain transfer agent (Figure S22). All other block copolymers solely show one diffusion coefficient independent from their composition. Further analysis such as transmission electron microscopy (TEM), small-angle x-ray scattering (SAXS) measurements are needed to clarify e.g. structural parameters as density differences and phase separation of the polymers to explain why only these block copolymers of this exact composition behave differently.

The probability of *racemic* linkages between single monomer units in the PHB blocks was between 74 and 78 % showing that block copolymerization did not influence the tacticity of the PHB block. In addition, ^{13}C spectra of PPDL-*b*-PHB were used to investigate possible transesterification reactions (Figure S18). Almost no transesterification reactions were present as there were no additional peaks and shoulders in the carbonyl region of PHB or PPDL. ^{1}H NMR spectra of the purified and dried PPDL-*b*-PHB were taken for determination of the PPDL/PHB ratios in the block copolymers (Figure S17). It was shown that the obtained ratios were in accordance with the monomer feed, thus, polymer compositions can easily be tailored by the monomer feed ratios. DSC was used for determination of the glass transition temperature and melting temperatures of the materials (Table S2, Fig. 3a). DSC curves of PPDL-*b*-PHB with different compositions retained the PPDL and the PHB melting temperatures which were not influenced by the other polymer unit. However, the T_m of PHB is more visible for block copolymers with higher PHB contents, whereas the T_g could not be detected for polymers with low PHB percentage. The DSC measurements indicated microphase separation of the two polymer phases.

To further study the crystallinity of the block copolymers, PXRD measurements were performed with the same samples that were studied

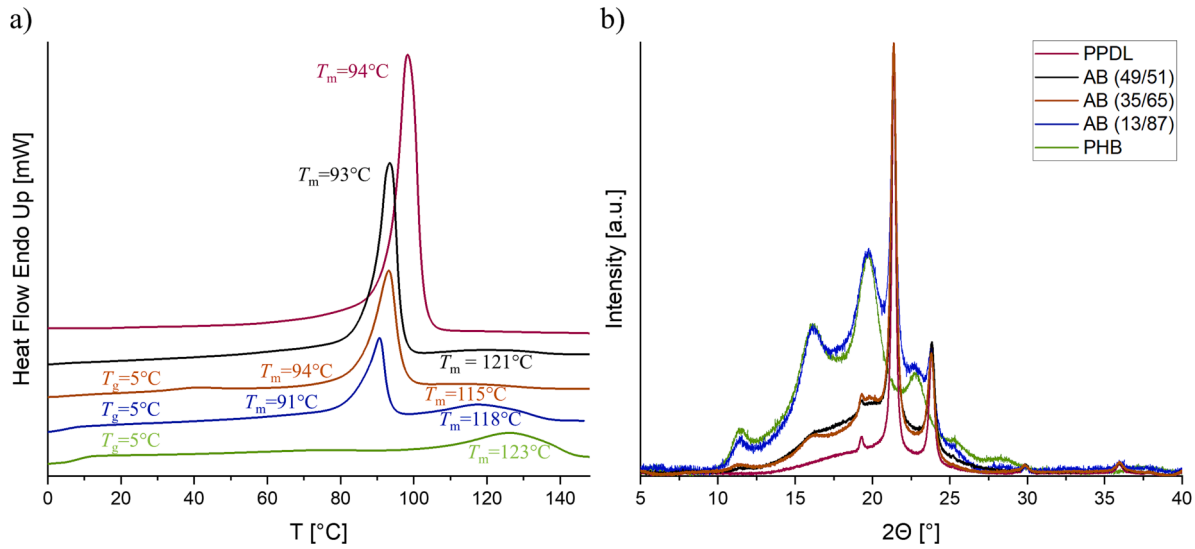


Fig. 3. A) Heating curves from DSC measurements of different samples. All curves show a section of the second heating cycle between 0 and 140 °C; b) Powder X-ray diffractogram of PPDL (pink; Table 1, Entry 5), PPDL-*b*-PHB (black; Table 2, Entry 3), PPDL-*b*-PHB (orange; Table 2, Entry 4), PPDL-*b*-PHB (blue; Table 2, Entry 5), PHB (green; $P_r = 0.76$; Table S1, Entry 3).

with DSC (Fig. 3b). PHB (green curve), and PPDL (pink curve) homopolymers as well as three block copolymers (black, orange and blue curve) of different compositions were measured. Reflexes of polymers that contained PPDL were normalized to the strongest reflex at 21.4°, caused by crystallinity of PPDL. PHB homopolymer (green curve) was used as a reference and normalized to the block copolymer with the highest PHB content (blue curve). PHB shows an underlying halo between 10 and 32° assigned to its amorphous regions as it is a semi-crystalline material. PHB diffractions of crystalline regions were found at 2Θ = 11.5°, 16.2°, 19.7°, 22.8°, 25.2° and 28.4°. Based on the diffraction of the pure PPDL (pink curve), its crystalline part was assigned to diffractions at 2Θ = 19.3°, 21.4°, 23.8°, 29.8° and 36.0°. In accordance with literature, the PPDL diffraction powder profile is characterized by two strong reflections at 2Θ = 21.31°, and 23.79° [65].

Due to its sharper diffractions, PPDL seems to be more crystalline than PHB but still shows a minor amorphous background between 12° and 27°. For the block copolymers, the amorphous background scattering is growing with increasing amount of semi-crystalline PHB. Simultaneously, the intensity of PHB diffractions is growing as well indicating that the crystal lattice of PHB is retained in all block copolymers. Due to the broad amorphous background of PHB, block copolymers do not show the reflex of PPDL at 19.3°. All other reflexes are sustained in the copolymers showing that also the PPDL lattice is sustained. These PXRD measurements show, that introducing PHB to PPDL maintains the PPDL and PHB crystallinity and introduces more amorphous contributions by amorphous PHB regions.

4. Conclusion

Herein, the first polymerization of biobased ω-pentadecalactone with [(ONOO)$^{t\mathrm{Bu}}$Y(bdsa)(THF)] was presented. ROP of PDL was performed at reaction temperature between 60 and 100 °C. Even high PDL equivalents of up to 200 could successfully be converted to high molecular weight polymers with molar masses up to 134.5 kg mol^{-1}. Introduction of *iso*propanol as CTA could convert the controlled coordination-insertion polymerization into an immortal ring-opening polymerization with full pentadecalactone conversions, already at a moderate temperature of 60 °C while having excellent control over molecular weight and molar mass distributions. Polymerizations were shown to be initiated by amido and alkoxide initiators. A polymerization temperature of 60 °C was chosen for block copolymerization studies in which the macrolactone PDL was combined with the small four-membered ring BBL. Sequential addition of the two monomers led to the PPDL-*b*-PHB AB block copolymers. SEC, DSC, and NMR measurements proved successful formation of the block copolymers and linkage of the two blocks. PXRD und DSC measurement were performed to study the microphase separation of the two blocks within the polymers and showed that the crystal lattices of both blocks are sustained within the copolymers. With the successful polymerization of PDL and its copolymerization with BBL to further tune material properties, this research is a successful approach to synthesize bio-derived polymers as a substitution for thermoplastic petrol-based plastics.

Declaration of Competing Interest

The authors have no conflicts of interest to declare.

Data availability

Data will be made available on request.

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Appendix A. Supplementary Data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2023.112449.

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